

AI-SOC-Assistant Evolution & Wins

A concise engineering brief showing how the project evolved from a Colab notebook into a modular, explainable SOC triage pipeline.

Capability	Professional Signal
5-class Random Forest detection	Classifies Normal, DoS, Probe, R2L, and U2R families instead of only binary anomaly status.
Isolation Forest fusion	Adds an unsupervised anomaly signal for traffic that does not match known attack-family patterns.
SHAP evidence bundles	Explains each flagged connection with top feature drivers and real network metric values.
LangGraph SOC analyst agent	Turns ML and threat-intel evidence into analyst-ready incident tickets.

Engineering Evolution

1. Feature Engineering: Robust Categorical Encoding

Migrated protocol, service, and flag handling from LabelEncoder to OneHotEncoder(handle_unknown="ignore"). This removes false ordinal relationships and keeps the live pipeline from crashing when an unseen service appears.

Win: Mathematically sound features and production-tolerant preprocessing.

2. Agent Orchestration: LangGraph ReAct Agent

Replaced legacy AgentExecutor patterns with langgraph.prebuilt.create_react_agent so the SOC analyst assistant can handle tool calls and stateful reasoning more reliably.

Win: Future-ready orchestration for dynamic threat-intelligence lookups.

3. Environment Constraints: Air-Gapped LLM Execution

Abstracted model initialization behind a provider factory and added Ollama support for local Mistral or Llama-style execution.

Win: Sensitive telemetry can stay inside restricted SOC environments.

4. Threat Intel Integration: Fault-Tolerant API Tools

Added bounded requests, timeouts, explicit exception handling, and compact NVD responses for AbuseIPDB and CVE lookups.

Win: External API issues become ticket context rather than pipeline crashes.

5. Explainable AI: Cross-Version SHAP Compatibility

Handled both legacy SHAP list outputs and newer multi-class array outputs when extracting the class-specific explanation vector.

Win: The explanation layer survives dependency-version changes.

6. Forensic Accuracy: Real-World Metrics

Routed unscaled feature values into the SHAP evidence bundle while keeping scaled arrays for model inference.

Win: Tickets cite analyst-readable packet, byte, and rate values rather than z-scores.

7. GenAI Guardrails: Ticket-Only Output

Strengthened the system prompt to require a strict incident-ticket structure, executable Splunk SPL blocks, and no raw SHAP float leakage.

Win: The LLM behaves as a translation layer from model evidence to SOC action.

Portfolio Positioning

The immediate polish pass adds a real MIT license, a runnable src/ pipeline, inspectable docs artifacts, pinned dependencies, and a sequential commit history. Together, those changes make the repo easier to clone, review, and discuss in interviews.